AI FOR EVERYONE:
EXPLORING THE POWER AND IMPACT OF ARTIFICIAL INTELLIGENCE

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Possible AI Topics

- **ChatGPT and its siblings**
- **Other important forms of AI**
- **How AI software “thinks”**
- History of AI development
- **Social and economic challenges of AI**
- **Deep questions about the nature of intelligence**
- Technical discussion of current AI research

ChatGPT and “LLM”s

- ChatGPT is one of a family of Large Language Models
  - Many LLMs available
  - ChatGPT (OpenAI), Claude (Anthropic), Grok (X), R1 (DeepSeek), Gemini (Google)
- Access is via simple, freemium websites
  - “Talk” in text-chat form with what seems like a very knowledgeable human, but is a program

Typical LLM Uses

- Comprehensive web search
- Summarizing and analyzing documents
- Generating or editing prose
- Tutorial discussions
- Brainstorming and recommendations

LLM Use “How To”

- It’s all about the prompt(s)
  - More than a question. Can include information, like documents, photos, etc.
  - LLM “knows” info only at the time of training, sometimes years earlier. Prompt gives current details.
  - Not just one prompt – engage in conversation. Entire conversation is “input” for each chat
  - Depth of remembered conversation varies by LLM – 1,000 words to 100,000
- LLM’s “hallucinate”, often very confidently
  - Check critical facts
  - Scan generated material for oddities

Example Uses

- Better web hunt – advice on furniture problem
  - Gives full description, various solutions, etc. (and it worked!)
- Document summary/analysis – IP agreement
  - Prompt includes entire document
  - Summarizes, points out issues, correctly evaluates
  - (But check for accuracy!)

Example Uses, cont..

- Editing prose – remove passive voice
  - Understands grammar, style
  - Example revision for PV is correct, succinct
- Brainstorming – name a cat
  - Sometimes offers silly suggestions with good ones
  - Will often come up with ideas you didn’t think of

Example Uses, cont..

- Write standard documents – reference letter
  - Prompt should include particular details, and general instructions
  - Can reuse prompt over and over w/ adjustments
  - Good at first drafts, but needs final polish (3rd paragraph, for instance)
- Tutorial Study – hierarchical Bayes' analysis
  - Gives good "textbook" summaries and explanations
  - Also response intuitively to questions
  - Wide ranging knowledge – like talking with universal graduate TA

Other AI

- Same principles of training and numerical patterns, but different designs and purposes.
- Diffusion Models
  - Generated movies in our intro
- Reinforcement Learning (RL)
  - Self-instructing. Requires automatically generable feedback.
  - Alpha Go example
- Physical AI
  - “Convolutional Neural Networks” or CNNs
  - Self-driving cars, robotics, face recognition

Example Uses – Other AI

- Artistic creativity – funny image
  - “Diffusion model”, e.g. DALL-E, Sora
  - Photos, short movie clips
  - May need feedback and adjustment, and may hallucinate
  - (Term “hallucination” originates with this type of AI)
- Generate audio of text **in your own voice**
  - Very convincing
  - Note that this means voices are not representative e.g. on phone calls

Higher “Reasoning” in LLMs

- Intuitively understanding higher math (math tutorial example)
- Writing and Revising a Story
  - Understands the context of the story
  - Summarizes; changes style
- Analyzing humorous images
  - “Gets” the humor
- Theory of Mind
  - Can empathize and visualize what’s in people’s minds

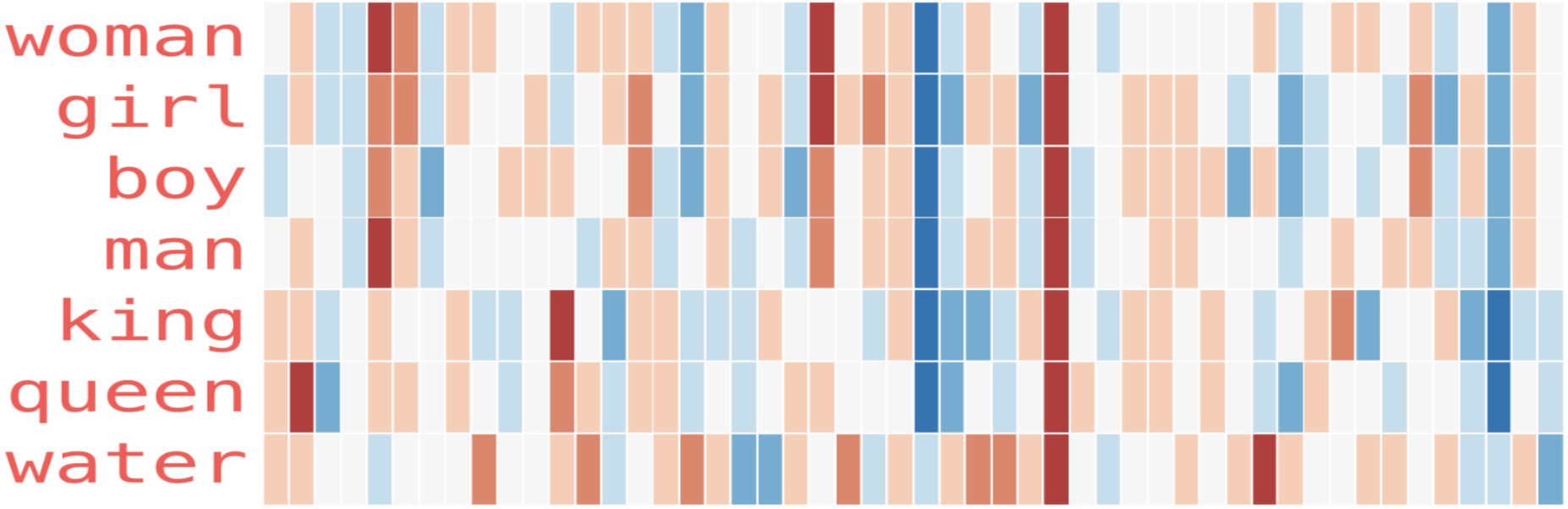
An LLM Is *Not*

- An improved search engine
  - Is able to web-search when useful, but…
  - … can work unplugged from the internet
- Hand-crafted code
  - No programming specific to the questions/answers
  - Could be written in one page of code – **12 equations**
- The whole modern AI story
  - Important other areas are making major advances too, e.g. “reinforcement learning”, computer vision

An LLM *Is*

- A sophisticated mathematical engine
  - 12 equations, but on “matrices” – huge grids of numbers
  - From 10 billion to over 1 trillion numbers
  - The “intelligence” is in those numbers (aka weights or parameters)
- Self-organizing!
  - The weights are “trained” by incremental adjustment
    - First, to match known texts
    - Then to adjust to human feedback given by RLHF
- Humans design the training process and the equations, but not the weights

Word/Idea “Embedding”



- This is how an LLM represents words, phrases and ideas
- Each word has a number sequence – an “embedding”
- But, with 12,800 numbers per word – many conceptual dimensions

How An LLM “Thinks”

- Starts with word embeddings representing words in your prompt and the current conversation.
- Combines words into higher 12,800-number patterns representing phrases, then sentences, etc.
- Predicts next “thought” as a number pattern and chooses closest next word.
- Repeat, word by word...

The Weights *are* the “Intelligence”

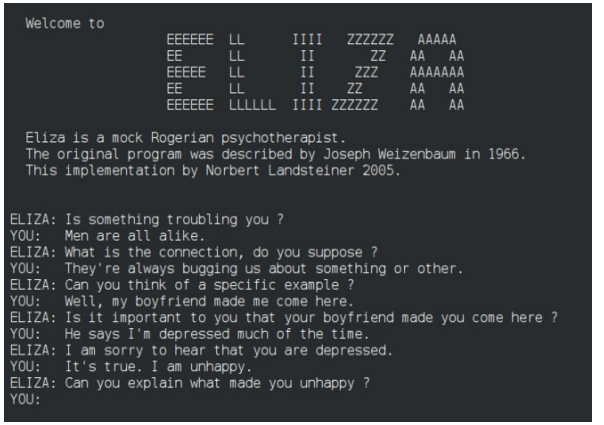
- All “reasoning” is by word/concept embeddings.
- And we (mostly) *don’t know* what each weight does.
- Could learn to converse in a language none of its programmers understand.
- Are world-models, theory of mind, etc. spontaneously emerging from the training?

Is an LLM Intelligent?

- “Yes” arguments
  - Vast possibilities in 1 trillion weights. Printed out they’d be 4 million large books.
  - Behavior suggests world-models, theory of mind, etc.
  - (But, we really don’t know since we’re not sure how the weights work in detail)
- “No” arguments
  - Just mindless numbers and 12 equations
  - Just predicting the next word.
  - (But, it’s improving *fast*, a human brain is just “mindless” neurons and synapses, and *how* is it picking that next word so intelligently?)

AI History 1950s - 1960s

- (1956) “Artificial Intelligence” first used as a term
- (1956) First good “theorem prover” – **symbolic AI**
- (1965) ELIZA basic “chatbot” in 200 lines of code
- (1960s) Hopfield, Minsky and others do early work in **“neural networks”**
- (1967) Minsky: “Within a generation … the problem of creating artificial intelligence will substantially be solved.”
- (1969) Minsky and Papert publish “Perceptrons”
  - Proves one-layer (all they knew how to train) neural networks very limited and triggers first **“AI winter”**



AI History 1970s-1990s

- (1970) Seppo Linnainmaa quietly publishes “backpropagation” algorithm, solving the multilayer training problem. *No one notices.*
- (1970s and 80s): Direct coding of logic in “expert systems”
  - Turns out to be limited and fragile
  - Vaunted “5th generation” computing fizzles
- (1986) Backpropagation algorithm (re)discovered.
  - Neural nets are back.
  - But training networks > 3 layers proves very hard. (Modern AI takes hundreds.)
- After brief burst of progress, a **second “AI Winter”**

AI History 2012-2024

- Four important advances lay ground for new neural network growth
  - more available training data via internet
  - computing power via GPUs
  - better neural network design: “activation functions” and architectures (neural net => deep learning)
  - vibrant and very intensive research community
- (2012) Neural networks, now “deep learning networks” are back, again!
- (2017) “Attention is all you Need” paper describes Transformers
- (2020-2023) Steady LLM improvement arrives at GPT-4

AI History – Big Ideas

- Early investigation of major ideas
  - Neural networks and “chatbots” since 1960s
- Alternating optimism and AI winters
  - Two major plateaus in progress
- Symbolic logic vs neural networks
  - The “bitter” (to some researchers) lesson – can’t program AI directly
- What’s different this time?
  - GPUs, massive training data, better algorithms, vibrant research community

Recent AI Events

- Nobel Prizes
  - Physics to John Hopfield and Geoffrey Hinton
  - Chemistry to the developers of Alpha Fold
- Improved LLM Architectures
  - MoE – “Mixture of Experts”
    - DeepSeek example
  - CoT – “Chain of Thought” – LLMs that reason
- Improvements in diffusion models
  - Sora movie generation

AI Events: late 2020s (?)

- Ongoing automation of white collar work
  - Writing, coding, art, music, law, healthcare, customer service, language translation, entertainment, etc.
  - My own experience: It's like a competent, widely read, and blindingly fast junior engineer.
  - May double or triple productivity of experienced professionals.
- Physical AI
  - AI with vision, movement, hearing
  - Robotics, self-driving cars, manufacturing, construction, etc.
- Scientific advances using AI, especially RL

Near Term Issues to Consider

- Legal questions of copyright and liability
- “Alignment” of AI with human values
- AI enables experienced professionals but replaces beginners creating “apprentice gap”
- Technological unemployment
  - Not like going from farm to factory to simple white-collar work
  - UBI might provide subsistence, but doesn’t provide purpose
- Which nation will lead? – Manhattan project of the 2020s

AI Future – Where is it heading?

- “… heavier than air flying machines are impossible”
  - Lord Kelvin, 1895
- Alchemistic Empiricism
  - Analogy of Chemistry in 1825
  - Results in fits and starts as we try things out by guess-and-by-golly
- Start with Nature, End with Design
  - Planes vs birds
  - Deep Learning vs brains : Geoff Hinton’s warning

AI Events: 2030s (??)

- Option A: AI progress will “plateau”
  - We are reaching limits on compute and on data
  - It’s plateaued before
- Option B: Continued progress reaching “AGI” or “superintelligence”
  - Unprecedented research community and infrastructure
  - Continued improvement in design and training procedures
  - Hardware is rapidly improving
- 5 years ago, many experts thought “A”. Now many think “B”.

Long Term Questions

- What can AI teach us about the nature of thought and intelligence?
- What if it becomes more intelligent than us?
- Hard Problem of Consciousness
- “Who is my neighbor?”

Advice to Students

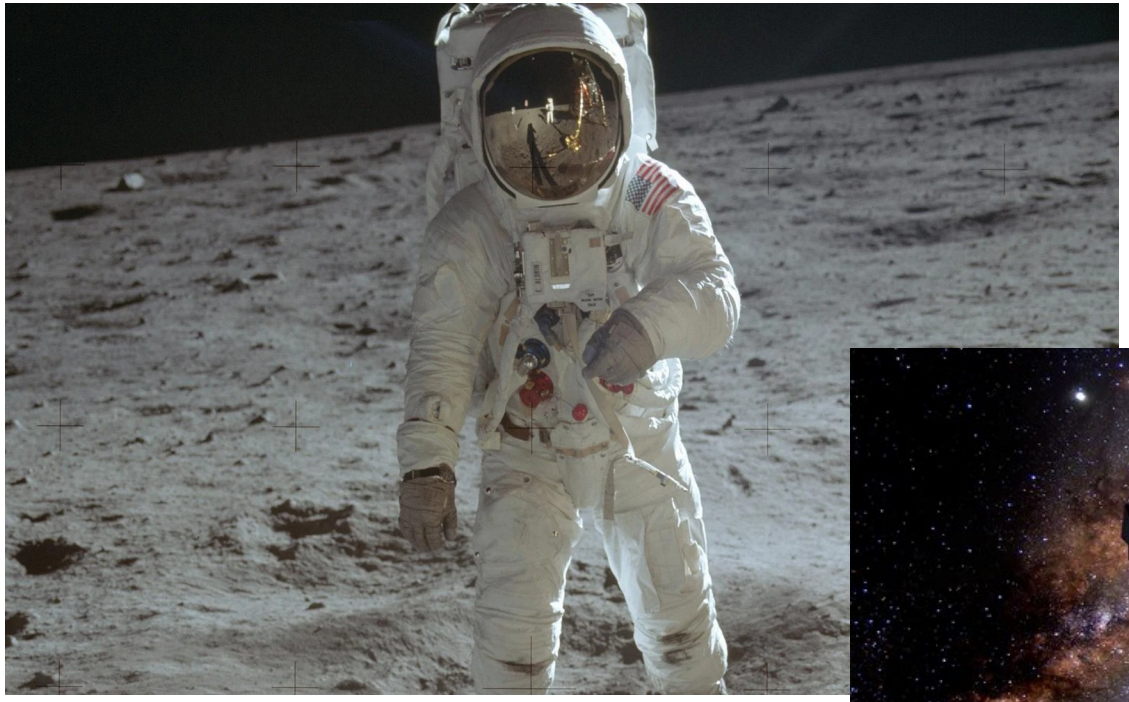
- Learn to create without AI first
  - Can’t be “in charge” if you are dependent
  - Gets you closer to content, like doing your own calculations
- Then, create using AI
  - Leverage AI to create more than you can alone
  - Use AI as “cointelligence” partner from whom you can learn as well
  - This is the workflow of the future

Future Predictions – Dystopian SF

- Premise: Technology is a power unto itself
- (1984) Blade Runner
  - LA is stygian hellscape in 2020s
- (1984) Terminator
  - LA is nuclear-blasted wreck with robots killing humans in 2020s
- (1970) Colossus
  - AI computer becomes world dictator in late 1970s

Future Predictions – CS Perspective

- Premise: Technology is Mind's creativity unfolding.
- "The mariner will have dominion over the atmosphere and the great deep .. the astronomer will no longer look up to the stars, – he will look out from them.." Science and Health p125: 25-29







Christian Science Perspective

- *“There is no life, truth, intelligence, nor substance in matter. All is infinite Mind and its infinite manifestation…” S&H p 468*
- *“Intelligence is omniscience, omnipresence, and omnipotence. It is the primal and eternal quality of infinite Mind...” S&H p 469*

Current Research – Optimization

- Make using AI less expensive after training
  - Reduce power consumption
  - Allow LLMs to run on edge devices like phones
- Not all weights are equally valuable; groups of weights can be compressed
- Precision of the weights can be looser, even down to one bit
- “Speculative Decoding”
  - Small “draft” LLM model does initial cut, confirmed by stronger model
- Knowledge Distillation (KD)
  - Train small LLM from “smart” larger one that offers nuance

Current Research – Intelligence

- Longer context windows
  - Squared complexity – requires creative selectivity in pairwise comparisons
  - Reasoning models
  - Architecture improvements: RoPE, better activation functions
  - Improved generalization
    - More data examples
    - RL-based training, especially in physical AI, coding, etc.
    - Double descent pattern and “grokking”

Current Research – Explainability

- Analyzing embeddings for concept patterns
  - Sparse autoencoders, and “golden gate bridge”
- Mechanistic Interpretability (MI)
  - Reverse engineering weight subnets “circuits”
- High Math Approaches
  - Mutual Information Theory
  - Sheaf Theory
  - Analysis of high-dimensional optimization surfaces, e.g. singular learning theory (SLT)